9 The Milky Way is the galaxy in which the Solar System is located.

(a) State what a galaxy is.

.....

..... [1]

(b) The Milky Way has a diameter that is approximately equal to 100 000 light-years.

Determine this distance in kilometres (km).

distance = ... km [2]

(c) Astronomers determine the speed and distance from the Earth of a far galaxy that is moving away from the Earth.

(i) State **one** observation that allows the speed at which a galaxy is moving away to be determined.

.....

..... [1]

(ii) State **one** different observation that is used to determine the distance to a far galaxy.

.....

..... [1]

(iii) State how the speeds of galaxies and their distances from the Earth are related.

.....

..... [1]

(iv) The best estimate for the Hubble constant H_0 is 2.2×10^{-18} per second.

Use this value to calculate an estimate for the age of the Universe.

age of the Universe = ... s [2]

[Total: 8]

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